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Non-Linearity of Skin Properties in Electrotactile Applications: Identification and Mitigation

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ABSTRACT Electrotactile displays can open a new sensory substitution channel to be utilized in a vast array of applications. But there are still challenges in different aspects of such systems. One of the most crucial concerns, is the effects of receptor fatigue. Our tests show that during prolonged exposure of receptors to the electrical signals, their sensitivity to the signal level changes gradually and adjustments in the signal's power are required to keep the receptors is the stimulated state. This was confirmed by monitoring the electrical current passing through the skin and calculating the corresponding impedance. More interestingly, the rates of the impedance changes are different for each part of the skin, indicating inconsistent rates of receptor fatigue for each region of the skin. These electrical properties of the skin were addressed in this research for the purpose of rendering consistent sensations for the users regardless of the person or skin conditions. To solve these challenges, two methods are employed: a voltage control system based on pulse-width modulation is used to adjust the signal power; and Kalman filtering is used to predict impedance changes in advance and supply the skin with the proper signal. The result is a self-contained automated system capable of managing the signal power for any user at any given time or skin condition.

INDEX TERMS Electrotactile display, electrical stimulation, sensory substitution, adaptive impedance mapping, skin electrical properties, electrostimuli, Kalman filter.

I. INTRODUCTION

An electrotactile display is based on the direct electrical stimulation of the skin for the purpose of transmitting information to the user. This can lead to the development of brain computer/machine interfaces (BCI/BMI).

Electrotactile displays can replace mechanical displays in many applications because of multiple advantages they have over the conventional mechanical approaches. One is the shorter refresh time that an electrical signal provides in comparison to the refresh time of a mechanical signal with various moving parts. The same effect can be seen on the skin. The response time of the skin to the electrical signal is significantly faster and has more accuracy [1], [2]. Other benefits include higher efficiency, flexibility, and convenience of the electrical stimulation [3].

Various applications have been suggested based on an electrotactile display [4]. The main concepts and mechanism of an electrotactile display have been investigated mostly by Bach-y-Rita *et al.* [5] and Kaczmarek [6]. Their work focused on a display to be placed on the tongue resulting in developing the *BrainPort* head-mounted system [7]. While most research in building an electrotactile display places the display on the tongue [8]–[10], other researchers have examined different locations on the body for an electrotactile display. These include the forehead [11], palm of the hand [12], [13], forearm [14], [15] , abdomen [16], and fingertip [17]. In our research, the display was placed slightly distal to the fingertip vortex. For the return electrode, a single contact is put on the palm of the hand [18]. There is an advantage for using a different location for the return electrode as to have a bigger space for the display. This approach has been previously used in other studies [19], [20]. This is in contrast to placing the return electrodes in a close proximity of the stimulating electrodes [21], which can result in limitations considering the two-point discrimination threshold (TPDT) of the skin [22] or electrical current leak to the surrounding electrodes [23].

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The final system is capable of reading any printed text and converting it to the Braille signal and sending it along side the directional signals to the fingertip.

One of the main challenges in building an electrotactile display is providing the appropriate signal considering the properties of the skin [25], [26]. A mapping approach can be used to mitigate these problems [27]. Other researchers used a normalized mean threshold map [28], [29], but our approach uses an adaptive spatial mapping, which has been extensively described in our previous work [30].

The frequency and intensity of the delivered electrical signal have an effect on the perceived sensation [31]. The ultimate goal is to select the proper sensation for the user to provide a comfortable feeling. One approach is to use electrical models to simplify the complicated arrangement of tactile stimuli while characterizing the properties of the skin [26]. The challenge is that the stratum corneum, the upper layer of the skin, has a significant effect on these models [25], which is subjected to different conditions such as sweat or other external factors because of the ionic conduction that distorts electric signals applied to the skin [32]. Also, research has shown that each point of the skin mandates a unique model of its own [33]. This has led researchers to employ pulse width modulation to control the signal for the contacts [34].

An increase of skin moisture levels during stimulation can result in current spikes [35] and is also not uniform across the skin, which can lead to reduced impedance at different rates. Also, the electroporation effect in the stratum corneum can lead to structural changes which forms small pores in the membranes in response to the applied electric signal [36]. Fortunately, this effect is reversible but can persist for several minutes. This means that a continuous change in the skin impedance for each point of contact which leads to complications of the controlling system as it is required to provide different signals for each point of the skin. In this research, we have examined these effects and have proposed an automated system based on the Kalman filtering to predict these changes. Also, using a pulse-width modulation we were able to provide each point of the skin with a unique signal. This resulted in a self-contained real-time control and adjustment of the signal for the purpose of supplying each point of the skin with the appropriate signal and same sensation.

This paper is structured into the following sections: in Section II, we discuss the methodology of the experiments and systems that are used in this study; section III explains the experiments being conducted in this work; in Section IV the proposed methods are presented; section V includes related discussion about the results and the proposed methods; and in Section VI we conclude the study.

II. METHODS AND EXPERIMENTS

A. HARDWARE AND SYSTEMS

The electrotactile display used in our research is built using 16 anode contacts where each is a copper circle with the diameter of 1.5*mm*. A sketch of the display with sub-array location numbers is shown in Fig. 1.

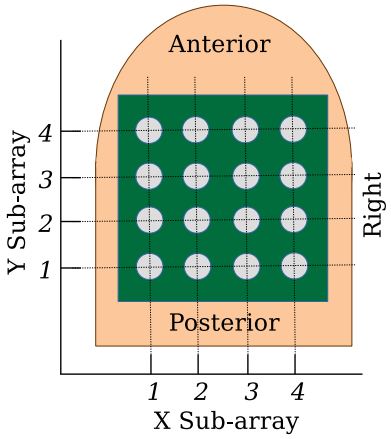


FIGURE 1. The regional areas of the electrotactile display are numbered for the analyses. Anterior to Posterior sub-arrays are referred to as *Y Sub-array* 1 to 4. Lateral sub-arrays are referred to as *X Sub-arrays* 1 to 4. The position numbers correspond to the voltage/current/impedance maps to follow.

The cathode return electrode, placed on the palm of the hand, has a diameter of 3*cm*. The reasoning behind this decision was explained in the introduction section. A switching board is also designed and built on top of a single-board computer (the Raspberry Pi 3) to control the status of the contacts. Fig. 2 shows the switching board and the electrotactile display.

The test signal used in our experiments has a frequency of 30*Hz* with a duty cycle of 10%, with a voltage below the detection threshold voltage level. These properties of the signal were chosen based on our prior study [37].

A block diagram of the system is shown in Fig. 3. An electrical current sensing circuit using a differential ADC based on the MCP3424 microchip was designed and utilized. The electrical current passes though a shunt resistor, the voltage drop across is sampled and amplified and passed to the ADC. The output is read by the program on the single-board computer. The program decides on the proper signal level based on the generated map and the Kalman filter estimations and predictions. The relays are used to send the signal to the electrotactile display. The relays are connected to a high voltage DC power supply (TTi PLH250). The power supply can generate any DC voltages ranging from 0 to 250 Volts with an accuracy of 0.1*V*.

Using pulse-width modulation, a pulse train signal with a very high peak-to-peak voltage is generated. The current output of the power supply is limited to 0.2*mA* in order to protect the subjects. The subject receives the signal on the skin and in the meanwhile an LED display shows the active contacts for testing purposes.

B. PARTICIPANTS

In this study, six participants (four males and two females) took part. They were all healthy graduate students and scholars at the University of Nevada, Reno. The average of age was calculated as 31 with the standard deviation of seven years.

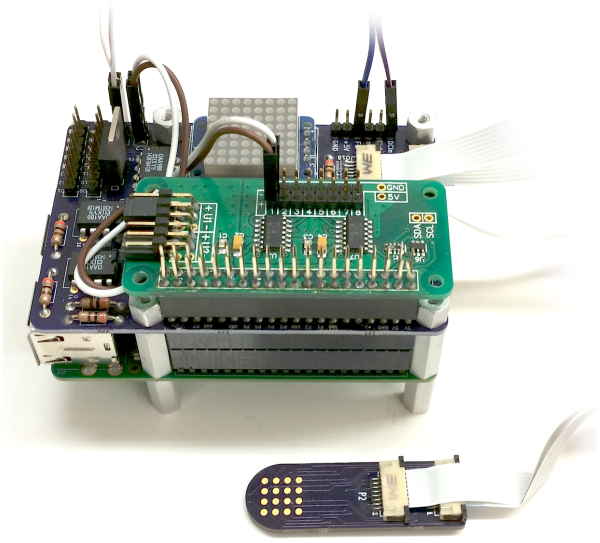


FIGURE 2. The electrotactile display fabricated as a 4 × 4 array of contacts is shown in the bottom of the picture. A switching board (the purple board) is designed and built on top of the Raspberry Pi 3. The ADC (the green board) is connected to the top of the switching board.

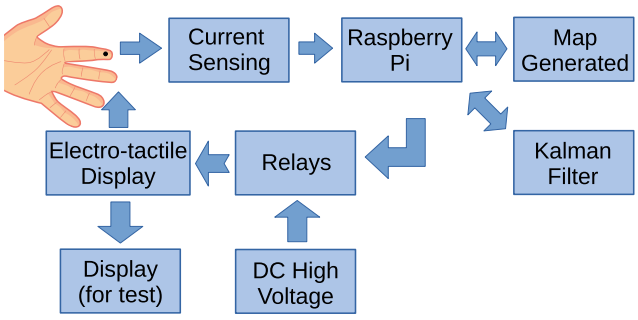


FIGURE 3. The block diagram of the system is shown here. The Raspberry Pi program receives the current measurements from the sensing circuits. Then, in coordination with the generated impedance map and the Kalman filter predictions, decides on the voltage level for each contact. The program then drives the relays that control the contacts on the fingertip.

There were no reports of any physical illness and they neither had any complications nor showed signs of any issues with their skin or fingers. The purpose and procedure of the tests were explained to each participant before beginning the tests. All of the contacts and the participant's fingertip and their palm were cleaned using Ethyl Alcohol 200 Proof and were left to dry.

The study was conducted in accordance with the protocols approved by the University of Nevada, Reno Institutional Review Board. Participation was with written informed consent.

For the experiments, a same constant voltage was applied to all of the contacts in sequences and the current passing through each one was recorded. The detection threshold voltage is the lowest voltage that would result in a sensation in the skin of the user. The constant voltage in these tests is set as a low value considerably below the detection threshold voltage so that the participants did not feel any sensations.

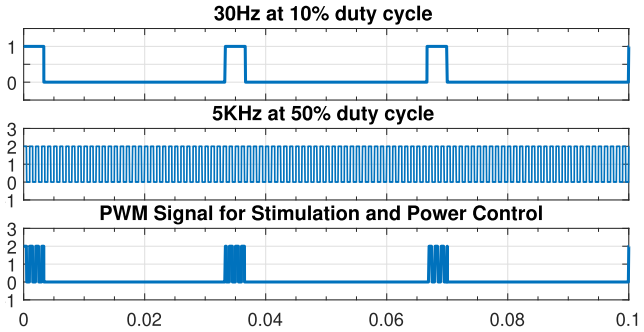


FIGURE 4. An overview of the PWM signal with the main signal of 30Hz and a modulation signal of 5KHz. This multiplication of these signals presents the opportunity to adjust the voltage levels through the program without the need of any external hardware.

There are several precautionary practices in place, such as limiting the current on the power supply and using a fuse in series to the circuit to make sure participants are safe.

C. VOLTAGE CONTROL USING PULSE-WIDTH MODULATION

One of the challenges in providing the skin with the appropriate signal is having the capability of adjusting the voltage delivered to the skin through the software. This is needed so that a manual adjustment of the voltage would be unnecessary. Adjusting the voltage through a software program can be achieved in various ways. Some approaches can be achieved by designing a software-hardware combination such as using switching systems. Other approaches can include using a DAC in connection with amplifiers. Each of these approaches has their own advantages and disadvantages. The most important disadvantages of these systems are that the final implementation will be bulky in size and weight. Especially since the voltage needed in our application to stimulate the skin can range from $30 - 100$ Volts DC. Any switching system or amplifier that is capable of handling such a high DC voltage will be large in size and the possible need of a temperature control system adds to its weight too.

Considering these challenges, an alternative method of controlling the voltage delivered to the skin was investigated in this research. The approach that was selected at the end was to use a multiplication of a significantly higher frequency signal with the normal signal that is delivered to the skin. This combination can be considered as a pulse-width modulation (PWM) signal and can be seen in Fig. 4.

The principle of our approach here is to essentially adjust the frequency and duty cycle of the PWM signal to change the voltage that is delivered to the skin. Since an electrical model of the skin consists of resistors and capacitors, this switching pulse train signal will be converted to a DC voltage as it enters the skin [38].

This approach was implemented in the program by using a PWM in conjunction with the main signal. The frequency and the dusty cycle of this signal can also be adjusted easily in the PWM subroutine. In this research, the frequency was kept at

5KHz (in comparison to the 30Hz signal that was delivered to the skin), and the duty cycle was changed from 0% to 100%.

There are multiple advantages in using this approach. The most important one is that the whole design is implemented in the software and no extra external hardware was needed. This helps with the size and weight of the final system, which is important since the end user needs to carry such as device. Another advantage of this approach is that any changes in the program are immediate and will take effect without any delays. This is helpful in adjusting the voltage instantaneously, followed by any changes in the conditions of the skin.

After implementing this approach, physical tests were done on users and the reports from them showed that the performance matched the intended design and the expectations. Using a low duty cycle signal, the users did not feel any sensations but as we increased the duty cycle, the signal reached a point that was distinguishable by the users and any further increases of the duty cycle resulted in an uncomfortable high signal sensation for the users.

One concern in this approach is that the system can only decrease the voltage and cannot increase it through the program. To address this, a normalized map of the impedance such as the one shown in Fig. 5 can be generated and used. In this approach, the contact that needs the highest amount of voltage is selected and a value of 1 is assigned to it. For every other contact, a scaled fraction is calculated.

The advantages of using this approach are that the required voltage for the maximum point is set as the input of the system, and the voltage control system can then adjust the voltage for all other contacts by using a fraction of the input voltage based on this map. This method was explained extensively in our previous paper [30].

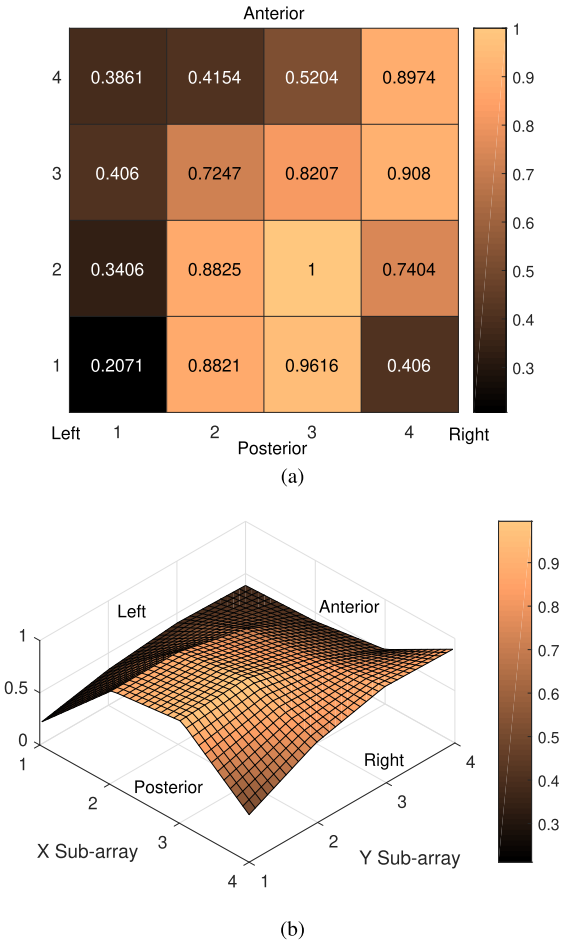


FIGURE 5. A normalized voltage map is generated by the system where the contact with highest impedance is given the value of 1 and every other contacts are scaled in relation to that contact. Figure (b) is the 3D map of the figure (a) for better illustration. The contact numbers follow the sub-array definition shown in Fig. 1.

D. REAL-TIME CURRENT MEASUREMENT

Measuring the current that goes through the skin is an essential factor in receiving the best sensation and updating the model and applied voltage. In our tests, it was determined that the comfortable tolerance range for the current is about $10\mu A$. This means that provided that the user is getting a comfortable sensation at $40\mu A$, the current can range from $35\mu A$ to $45\mu A$. Any currents less than this range will not provoke any sensations for the user and any currents higher than this range will cause an uncomfortable sensation or even pain.

Since the acceptable range for the current is very narrow, a highly accurate and efficient mean for measuring the current is needed to guarantee the user is not receiving any current spikes outside of that specific range. To achieve this, a differential ADC based on the MCP3424 microchip was used. The current passes a shunt resistor and the voltage drop across is sampled, amplified and passed to the ADC.

After implementing this design, it became clear through the tests that this setup is capable of measuring the current passing through the skin with very high accuracy. The only area that it can benefit from further improvements is the speed of the sampling, which is a few samples per seconds at this moment due to the processing capabilities of the ADC and the single-board computer.

III. EXPERIMENTS AND RESULTS

A. THE TEMPORAL CHANGES OF THE IMPEDANCE OF THE SKIN

At this point of the research and by having access to the measurements of the electrical current through the program, multiple tests were performed to monitor the current amount that goes through the skin during the stimulation. It immediately became clear that the current that goes through the skin changes as a function of the time. This means that the current never reaches a constant level and fluctuates very frequently. Having the input voltage fixed, this implies that in fact the impedance of the skin changes as a function of time. This was a novel finding as it was not addressed in the literature before. The skin fatigue is a known condition that effects the impedance but it does not account for the level of the fluctuations we observed in the tests.

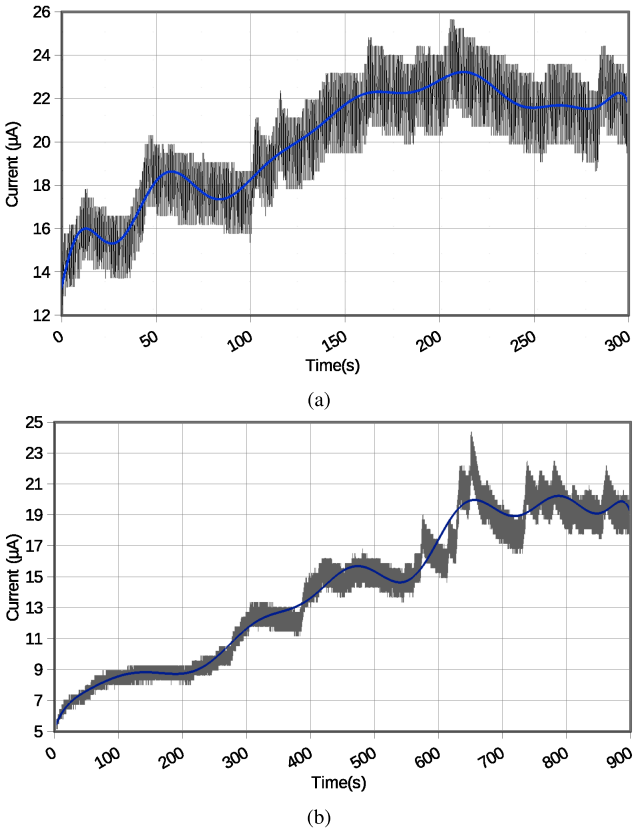


FIGURE 6. (a) A five-minute test shows that the current increase over time rejects the hypothesis of an electrical current plateau. (b) A 15-minute test shows that the current increase over time can exceed the expected limits. In these tests, the voltage is kept at a constant level and the blue line is a running average fit of the measurements. The absence of a current plateau indicates a continuous decrease in skin impedance.

In our experiments, it became noticeable that in most cases, the current passing though the skin would increase initially but may soon reach a plateau. But, allowing the current to pass through the skin for an extended time revealed that such a saturation point may not ever be reached. Fig 6-(a) and (b) show this characteristic.

The longest time that this test was run was for 15 minutes. This is shown in Fig. 6-(b). In this specific test, the current starts at about $5\mu A$ and increases with some fluctuations over a 15-minute period time. At the end, the current has reached to almost $20\mu A$. This shows the current being quadrupled over 15 minutes. Such an enormous increase in the current shows a very dramatic decrease of the skin impedance over time. The voltage in all of these tests were kept at a constant level throughout the tests. This decrease in skin impedance suggests that the formation of the paths in the skin, where current passes through, changes as a function of time.

It is important to note that in regard to the decrease of the skin impedance (or the increase of the current), given a reasonable amount of time to rest, the skin will restore to its initial characteristics. This means that even after subjecting the skin to 15 minutes of current passing through, if the circuit becomes disconnected and a reasonable amount of time passes, at the start of a new test, the skin behaves as if such a dramatic increase has never happened. This means the skin characteristics are fully reversible.

It is worth mentioning that the maximum time duration that these tests were run was 15 minutes and a longer duration of time was not investigated in this research. The reason for this time limit is the fact that any pressure changes of the fingertip would change the structure of the skin under stimulation and consequently change the level of the current passing through. Therefore, the fingertip needs to be kept at an exact position throughout the tests. Tests longer than 15 minutes failed because the users showed signs of fatigue and unintentional muscle movements happened resulting in inaccurate current readings.

B. THE SPATIAL CHANGES OF THE IMPEDANCE OF THE SKIN

Based on the results of the previous section, we were able to measure the changes in the fingertip skin impedance for each of the 16 contacts as defined in Fig. 1. The results of tracking the current based on a constant voltage is shown in Fig. 7. The Figure 7-(a), is a 3-minute test of the current measurement of all of the contacts of the electrotactile display given a constant voltage of $30V$. The Fig. 7-(b) shows the same test using a voltage of $40V$ and for a duration of 7.5 minutes.

The fascinating finding in these tests is that the impedance of the skin is different for each part of the skin and more importantly, there is a grouping characteristic based on the impedance. The plots in these two figures are drawn using four groups of colors to distinguish these groups. Fig. 8 shows the grouping colors.

The highest current, shown in the purple colors, are the four contacts in the middle posterior. In every test, these four contacts have received the highest level of the current throughout the test. The next group, shown in gold/brown colors, are six contacts to the top and on either side of the previous group. Next group, shown in green colors, are four contacts in the most anterior and again to the either side of the previous group. Finally, the last group, shown in blue colors, are the two contacts in right and left anterior positions.

IV. PROPOSED METHODS

A. SELF-GUIDED REAL-TIME ADJUSTMENTS OF THE VOLTAGE

Considering the characteristics of the skin that were described in the previous section, it is clear that a self-guided real-time system is needed to decrease and adjust the voltage automatically over time as the impedance of the skin drops. This will help with maintaining a steady electrical current to the skin to keep the sensation the same despite the fact that the internal characteristics of the skin may have changed significantly.

To achieve this goal, a straightforward feedback system was designed and implemented to increase or decrease the voltage level to keep the current the same. The changes of the voltage are applied by changing the PM signal described before. Fig. 9 shows the result of this system.

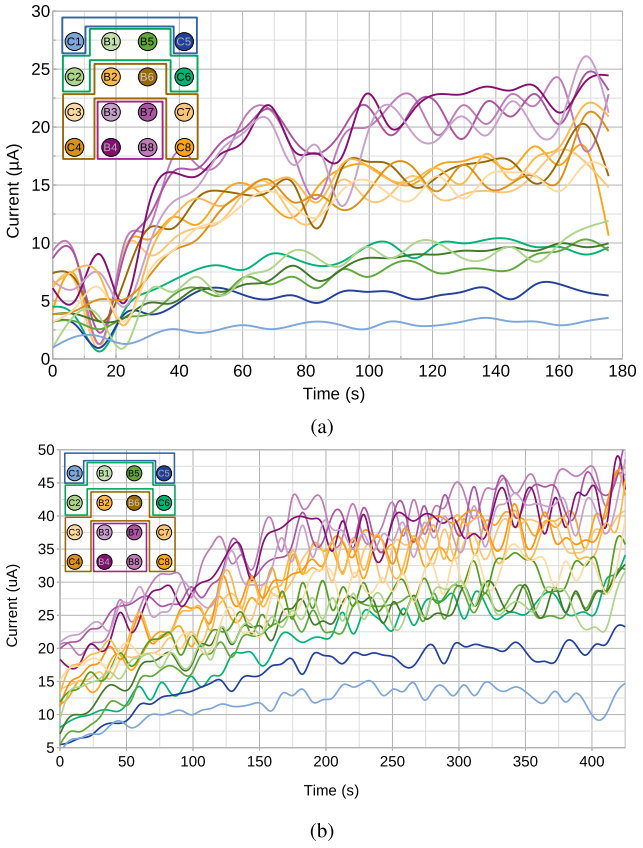


FIGURE 7. A 3-minute test (a) and a 7.5-minute test (b) are performed on all contacts of the electrotactile display. The results show that four groups can be identified based on their location and their current levels that passed through during the tests. A zoom-in picture of the legend is shown in Fig. 8. The test (a) is run at 30*V* and test (b) is run at 40*V*. That difference is seen in the different current levels.

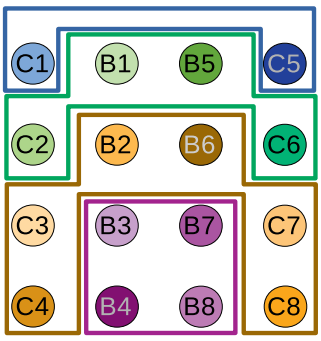


FIGURE 8. The grouping that was found in the previous tests can be seen here. This map follows the sub-array numbering in Fig. 1. In this picture, from bottom to top, the purple group has the highest current. Next is the gold/brown group. The next group is the green colors. Finally, the blue colors show the group with the lowest current.

As it can be seen in Fig. 9, the voltage started at about 30% and over a 4-minute period time had to be increased to 80% and finally stabilized at about 60%. These adjustments are done in the program automatically to regulate the electrical current levels.

The purpose of this experiment was to find out whether adjustments of the voltage is sufficiently enough to keep the current and the sensation at the same level over an extended

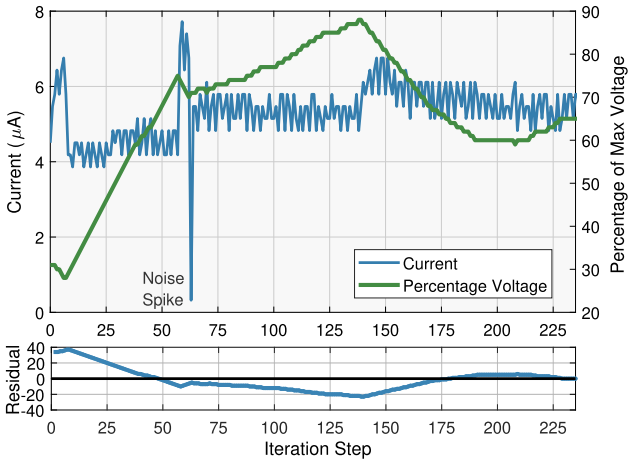


FIGURE 9. An automatic feedback system (shown in the green line) adjusts the voltage level to regulate the current (shown in the blue line) despite changes in impedance.

period of time. It became clear that considering the range of adjustments of the voltage, such a scenario can be implemented and the test results showed that the sensation was kept at the same level too.

B. USING KALMAN FILTER TO PREDICT THE CHANGES IN IMPEDANCE

As we saw in the previous sections, the current that goes through the skin changes as a function of time, but these changes are not very predictable. In an ideal situation, a model could be obtained for the changes of the current and future adjustments could be planned based on that model. Based on the experiments though, the changes in current do not follow a distinguishable pattern or model. To solve this problem, a Kalman filter was used to predict the future values of the current based on the measurements of the current at any moment:

$$\hat{x}'_k = \hat{x}_k + K'(y_k - H_k\hat{x}_k) \quad (1)$$

$$P'_k = P_k - K'H_kP_k \quad (2)$$

$$K' = P_kH_k^T(H_kP_kH_k^T + R_k)^{-1} \quad (3)$$

where y_k represents the electrical current measurements and $\hat{x_k}$ represents the predicted future electrical current values.

This method was implemented in the program and was combined with all of the advancements described in the previous sections. That is, the current is measured in each loop, next a Kalman estimate is calculated for the future step, then a voltage adjustment based on the future estimate is calculated and applied using a PM signal, and finally the new current is measured and the Kalman filter updates the estimate in the next loop. The complete loop can be seen in Fig. 10.

Two results of implementing this method are shown in Fig. 11 where it can be seen that the Kalman filter perfectly predicts the changes in the current and the algorithm changes the voltage (shown as the green line) to adjust and regulate the current levels. In these figures, note the constant level

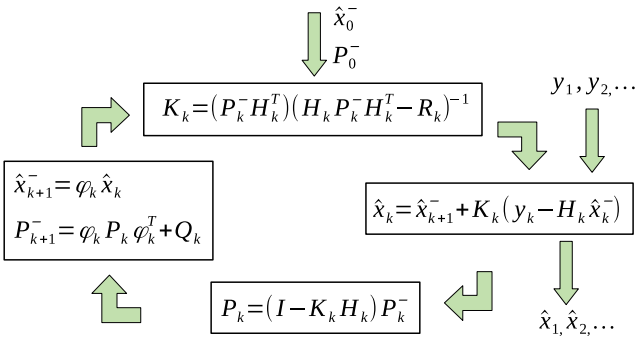


FIGURE 10. A Kalman filter feedback loop is shown. Here, y represents the current measurements and $\hat{x}$ represents the predicted future current.

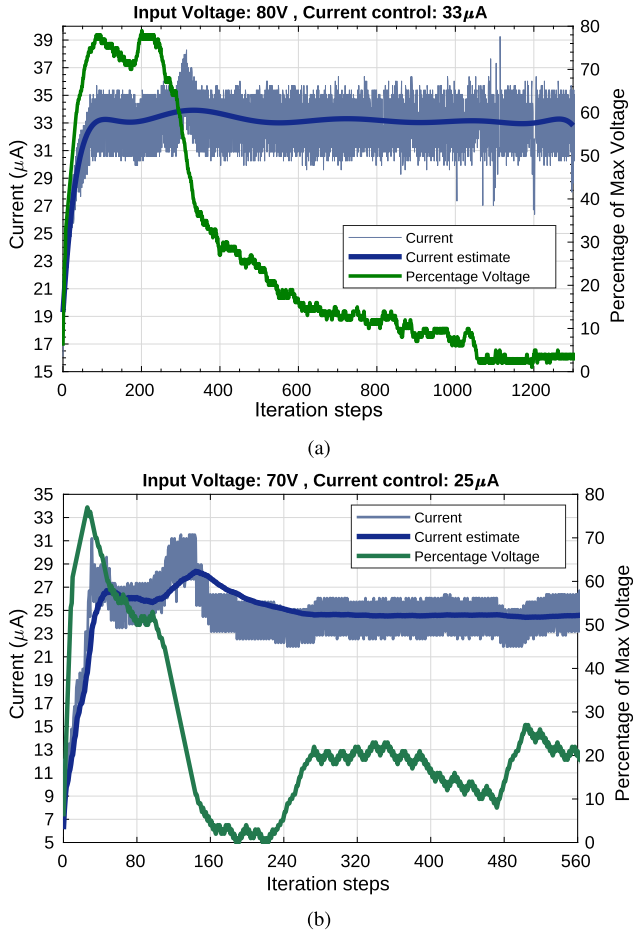


FIGURE 11. These figures show two different results, from two separate tests, of the Kalman filter prediction (the dark blue color) and adjusting the voltage accordingly (the green color) to keep the current measurements (the light blue color) at a constant level. Both tests were run on the same user.

of the current at the end of each experiment. Changes of the voltage ranged from almost 80% to about 15% at the end of the experiment in Fig. 11(a).

In any of these experiments, the Kalman filter was able to successfully predict the changes in the current and the adjustment of the voltage resulted in regulating the current at a constant level by the end of the experiment. This is a solution to the problem of regulating the current since the changes of the voltage shows that the decrease of the impedance of the skin is not following a predicable model.

One limitation in this implementation is that the range of the changes of the voltage is limited and may exceed the lower bound of the signal limits at some point. It is not clear how much drop in the impedance will occur if the system is used for an extended duration of time such as hours. The reason behind the lack of experiments in this area was explained in section III-A.

V. DISCUSSION

The interactions of the electrical current and the internal structure of the skin is important and challenging because the end goal of the electrotactile stimulation is supplying the user with a "comfortable" sensation. The real challenge in this case is that the comfort level is a subjective measure and cannot be quantified quantitatively. This leads to various efforts in providing an appropriate signal for the user and applying continuous corrections to ensure and guarantee the sensation feeling.

Our approach has been successful in this regard because of our consideration of multiple factors. The most noticeable consideration is our effort for gather an extended range of real-time measurements and to use prediction methods to automatically adjusting the signal properties hundreds of times in every minute. This rapid and continuous updating has provided the opportunity to better characterize the skin responses and adopt the proper adjustment in real-time.

The experimental results presented in the previous section demonstrate that our efforts have been successful in regulating the electrical current and the subsequent signal. The self-guided approach that was implemented, ensures this happens automatically. It is also important to note that these calculations and adjustments happen in real-time using a low-power single-board computer. One of the reasons behind using a Kalman filtering in this case was the match between the level of the calculation required and the computational resources available. Other approaches were also tried but did not satisfy these measures.

One interesting observation from the selected results presented in Fig. 11, is that the voltage level adjustments (the green plot) are not predictable either. Two results were selected and presented here to highlight this finding. Fig. 11(a) shows a continuous decrease in the voltage level for the purpose of regulating the current, but Fig. 11(b) shows that this was not needed in this case. It is worth mentioning that both of these results are from tests done on the same user.

It is Note that the difference between the iteration steps is not an indication of time duration but is simply a matter of Kalman filter calculations.

Explaining or justifying these differences, especially considering the fact that the user was the same person for both experiments, is not easily possible. The fact that can be considered is that the skin conditions can follow multiple

unrelated factors such as the hydration level, rate of sweating, and other factors. Even the possibility of the effect of the pulse rate was considered but could not be confirmed through our experiments. This vast array of these naturally unrelated parameters can influence the skin condition and the impedance levels. Tracking and analyzing all of these factors are beyond the scope of this research and require its own specific investigations.

Reasonable alternative is to consider the skin conditions as a black box with unknown characteristics and only rely on the exhibited effects that can be measured and applied to the skin. In this regard, the research presented in this work demonstrates substantial advancement with proper and outstanding outcomes.

VI. CONCLUSION

This work presents our new findings and improvements to the electrotactile displays built in our previous research. The most important finding in this research is the temporal and spatial changes of the impedance of the skin. Through our experiments, we found that the skin impedance level can change dramatically over even a short period of time. Also, these changes can be seen for each point of the skin, and more interestingly, the rate of the change may not be the same for each point. This was confirmed even for contacts points that were in the close proximity of each other. This led to categorizing these contact points into four groups, with each group exhibited similar characteristics. The location of these groups also followed our previous hypothesis.

Based on these findings, a signal regulation system was proposed and implemented to adjust the signal properties automatically and in real-time. A self-guided method was presented based on pulse modulation to adjust the pulse train and subsequently the voltage level. This modulation is based on a multiplication of two signals with different frequencies and duty cycles. The results show a perfect automatic voltage adjustment.

Furthermore, the Kalman filtering was employed as a prediction and estimation method for the changes in the impedance and the current levels. The Kalman calculations were adopted perform in real-time on a single-board computer with low power resources. The results show an excellent tracking and prediction of the current. The Kalman filter results were used as feedback for the signal control procedure described above.

By combining these methods, we introduced an automated mechanism that was capable of measuring and tracking the current passing through the skin and performing estimations and predictions. Moreover, the system is capable of using this information for adjusting and regulating the electrical current levels even when the skin impedance had dramatic changes. This signal regulation results in a constant feeling sensation for the purpose of the electrotactile stimulation.

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CONFLICT OF INTEREST STATEMENT

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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